

UNIVERSITY OF CALIFORNIA
SANTA CRUZ

**LLM-COORDINATION: DEVELOPING COORDINATING
AGENTS WITH LARGE LANGUAGE MODELS**

A thesis submitted in partial satisfaction of the
requirements for the degree of

MASTER OF SCIENCE

in

COMPUTER SCIENCE AND ENGINEERING

by

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December 2023

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2023

Table of Contents

List of Figures	v
List of Tables	vii
Abstract	viii
Dedication	x
Acknowledgments	xi
1 Introduction	1
1.1 Document Structure	4
2 Background	6
2.1 Autoregressive Large Language Models (LLMs)	6
2.2 Large Language Model Agents	7
2.3 Multi-agent Coordination	8
2.4 Theory of Mind	9
3 Related Work	10
3.1 Multi-agent Coordination	10
3.2 Planning and Reasoning with Large Language Models	11
4 Methodology	13
4.1 LLM-Coordination Framework	13
4.1.1 Game and Layout Description	15
4.1.2 State Representation	15
4.1.3 Feasible Action Generation:	17
4.1.4 The Large Language Model	18
4.1.5 Action Manager	18

5	Evaluation Environments	21
5.1	Collab Capture	21
5.2	CollabEscape	22
5.3	Overcooked	22
5.4	Overcooked-Explicit Assistance	23
5.4.1	Gated Delivery	24
5.4.2	Locked	24
6	Experiments	25
6.1	Theory of Mind Inference and Situated Reasoning	25
6.2	Sustained Coordination	26
6.3	Robustness to Partners	27
6.4	Explicit Assistance	27
7	Discussion	28
7.1	Theory of Mind Inference and Situated Reasoning	29
7.1.1	Results	29
7.2	Sustained Coordination	30
7.2.1	Baselines	30
7.2.2	Results	31
7.3	Robustness to Partners	33
7.3.1	Results	33
7.4	Explicit Assistance	34
7.4.1	Results	34
8	Case Studies	37
9	Limitations	41
10	Conclusion	42
	Bibliography	43
A	Prompt Details	53
A.1	Overcooked Game Description Prompts	53
A.2	Overcooked Layout Prompts	54
B	Details of LLMs used in Experiments	58

List of Figures

2.1	A summary of LLM Agents and their components [39]	7
4.1	Visual summary of the LLM-Co framework. Our framework serves as the backbone for an individual agent, focusing on bringing out its coordination ability. The framework translates abstract game details into an LLM-compatible format and then utilizes the generated LLM output to take actions in the game world.	13
5.1	The CollabCapture game involves two agents, Alice (Blue) and Bob (Green), chasing a thief across multiple rooms. Some rooms are connected by doors, which can be controlled by buttons in different rooms.	21
5.2	All layouts from the overcooked environment we use for our tests. The two agents Alice (Blue) and Bob (Green) need to collaborate to cook, plate, and deliver onion soups. From Left to Right: Cramped Room, Asymmetric Advantages, Forced Coordination, Coordination Ring, and Counter Circuit.	22
5.3	Additional Layouts that require agents to explicitly help their partner complete a delivery. These new layouts utilize walls and gates to create situations requiring explicit assistance.	23
7.1	LLMs performance on the LLM-ToM-Reasoning test set. Partner action intent prediction accuracy shows the Theory Of Mind ability of LLMs under test and the optimal action reasoning accuracy infers the Situated Reasoning effect of LLMs under test. GPT-4 achieves the best performance among tested LLMs.	28
8.1	The LLM-Co agent (Blue) understands that its partner has already picked up an onion and will probably end up placing it in the cooker to complete the requirement. So, it chooses to "pick up a plate" which would help in the next stage.	37

8.2	The LLM-Co Agent (Blue) thinks about the partner’s mind states (shown in red) and makes an informed decision (highlighted in green) to move away to make way for the partner agent)	38
8.3	The LLM-Co Agent (Blue) realizes that the onion it picked up cannot be used and decides to correct itself by placing it on the kitchen counter to free up its hands.	39
8.4	The LLM-Co Agent (Blue) recognizes that its partner is stuck behind a closed door an chooses to open the gate g0 to enable them.	40

PREVIEW

List of Tables

4.1	Complete medium-level action set.	20
7.1	Comparison of game play between self-play baselines (PPO, and PBT) and LLM-Co Agents. LLM-Co agents outperform RL methods on 4 out of 5 layouts, demonstrating highly effective reasoning under sustained coordination.	31
7.2	Comparison of AI-Human Proxy Game play. We compare Behavior Cloning Agents, PPO_BC Agents with LLM-Co agents utilizing the GPT-4 LLM. The LLM-Co agents are able to outperform or match the performance of Reinforcement Learning models, indicating that LLM agents are robust to the choice of partner agents.	33
7.3	Comparison of Gameplay in the Overcooked-Assistance Layouts with and without Helper Directive. The results indicate that the Large Language Model needs to be prompted to be aware of situations where their partner might need assistance in order to be effective in the Overcooked-Assistance layouts.	34
7.4	Comparison of Gameplay on Overcooked-Assistance Layouts between RL baselines and LLM Agents. The RL baselines being able to effectively solve the deliveries indicates that the environments are solvable through self-play training. The high scores achieved by LLM agents demonstrate that LLM agents are capable of reasoning for providing explicit assistance to their partners.	35

Abstract

LLM-Coordination: Developing Coordinating Agents with Large Language
Models

by

Saaket Agashe

It is essential for intelligent agents to not only excel in isolated situations but also coordinate with partners to achieve common goals. Current Multi-agent Coordination methods rely on Reinforcement Learning techniques to train agents that can work together effectively. On the other hand, agents based on Large Language Models (LLM) have shown promising reasoning and planning capabilities in single-agent tasks, at times outperforming RL-based methods. In this study, we build and assess the effectiveness of LLM agents in various coordination scenarios. We introduce the LLM-Coordination Framework to enable LLMs to complete coordination tasks. We evaluate our method on three game environments and organize the evaluation into five aspects: Theory of Mind, Situated Reasoning, Sustained Coordination, Robustness to Partners, and Explicit Assistance. First, the evaluation of the Theory of Mind and Situated Reasoning reveals the capabilities of LLM to infer the partner’s intention and reason actions accordingly. Then, the evaluation around Sustained Coordination and Robustness to Partners further showcases the ability of LLMs to coordinate with an unknown partner in complex long-horizon tasks, outperforming Reinforcement Learning baselines. Lastly, to test Explicit Assistance, which refers to the ability of an agent to offer help proactively, we introduce

two novel layouts into the Overcooked-AI benchmark, examining if agents can prioritize helping their partners, sacrificing time that could have been spent on their tasks. This research underscores the promising capabilities of LLMs in sophisticated coordination environments and reveals the potential of LLMs in building strong real-world agents for multi-agent coordination.

PREVIEW

To My Parents and my Partner.

PREVIEW

Acknowledgments

I would like to express my deepest gratitude to Dr. Xin Eric Wang, my thesis supervisor, for their unwavering support and guidance throughout the course of this research. Your expertise and insights have been invaluable to my work.

I am also immensely grateful to Professor Yi Zhang and Professor Ian Lane, whose expertise and feedback significantly contributed to the refinement of this thesis.

Special thanks to my colleague Yue Fan for his invaluable insights and suggestions regarding this project.

I am indebted to my family and my partner for their endless love, moral support, and encouragement. Your belief in me has been a constant source of strength and motivation.

Lastly, I extend my appreciation to the staff and faculty of the CSE Department at the University of California, Santa Cruz, for providing a conducive environment for learning and research.

Chapter 1

Introduction

The need for coordination can arise from mundane tasks like cooking and cleaning to more essential tasks like coordinated search-and-rescue operations. The assimilation of AI tools and agents into the human experience cannot be complete without AI agents being able to coordinate in a similar manner. Current methods for Multi-Agent coordination predominantly utilize Reinforcement Learning approaches that suffer from issues of sample efficiency, lack of robustness to new partners, and lack of interpretability. More Recently, Agents based on Large Language Models have emerged as a new paradigm for developing situated, task-oriented agents [36, 42]. LLMs are capable of solving reasoning problems with minimal downstream training data due to their pretraining and generate free-text explanations, making them promising candidates for developing coordination agents. While the planning and reasoning abilities of Large Language Models for common-sense and logical reasoning tasks have been extensively studied, their emergent cooperative reasoning abilities are underexplored.

In this study, we introduce the LLM-Coordination Framework to facilitate multi-agent coordination using Large Language Models. We will refer to agents using the LLM-Coordination framework as LLM-Co agents. The LLM-Co framework is an Agent framework with the following features

- Programmatically converts real-time state information to textual description that the LLM can parse.
- Generates feasible action space based on the current state.
- Parses LLM response and converts it to a sequence of low-level actions
- Maintains a memory of previous actions for the LLM

We then systematically evaluate the LLMs and the LLM-Co framework on the following critical competencies required for effective coordination: Theory of Mind Inference (understanding others' beliefs and intentions), Situated Reasoning (contextual action analysis), Sustained Coordination (long-term action planning and adjustment), Robustness to Partners (adapting to new collaborators), and Explicit Assistance (actively aiding partners).

We begin by conducting a preliminary study that allows us to determine the baseline abilities of various Large Language Models and their suitability to be used in the LLM-Coordination Framework. We do this by examining LLMs' Theory of Mind Inference and Situated Reasoning capabilities, which are foundational for coordination. These abilities enable models to understand others' intentions and contextualize these

understandings within their environment. We introduce the "LLM-ToM-Reasoning Test Set," comprising diverse scenarios from multiple coordination environments, to assess LLMs' performance in deducing optimal actions based on their partner's intentions and environmental context. Comparing four different LLMs (GPT-4, GPT-3.5-turbo, Vicuna-33B, and Vicuna-13B), we find GPT-4 to be significantly superior, getting an almost perfect score. The other LLMs cannot provide sufficient correct responses to be suitable for following multi-turn coordination.

Next, we evaluate the sustained coordination abilities of LLM-Coordination Agents. We use GPT-4 as the LLM of choice as it is the only candidate that provides acceptable ToM and Situated Reasoning scores. Providing the LLM with Task Description, Turn-by-turn state information, action memory, and a list of feasible actions at each timestep, we explore the sustained coordination reasoning in LLMs on the Overcooked-AI benchmark. We compare the performance of LLM Agents (w. GPT-4) with Reinforcement Learning (RL) based baselines, which are the gold standards for AI-AI gameplay. We then experiment with varying the partners in the Overcooked Environment. We pair LLM-Co agents with proxy human agents to test the agents' robustness to partners. We observe that LLM agents perform better than or equal to the RL baselines in both AI-AI and AI-human proxy gameplay.

Previous works introducing novel methods for developing Collaborative Agents [46, 21, 45] using LLMs do not test their abilities to understand the concept of common payoff and proactively help their partners (Explicit Assistance). We introduce two new layouts in the Overcooked environment, including a gate element that forces agents

to assist their partners in completing deliveries. Through experiments on these new layouts, we discover that LLM agents can determine the right strategy needed to help out their partners. However, they require a "helper prompt" to be attentive to situations where their partner may need such help. We train RL baselines on these new layouts and discover that LLMs with helper directives outperform these baselines.

This research highlights the potential of LLMs in coordination reasoning and contributes a novel perspective to the field of AI-agent collaboration. The LLM-Coordination Framework opens new avenues in the development of AI agents capable of sophisticated, context-aware, and coordinated decision-making, setting a foundation for future explorations in multi-agent AI systems.

1.1 Document Structure

- Chapter 2 will cover some prerequisite concepts and terminology about LLMs, LLM Agents, and Multi-agent Coordination that will be used throughout the document.
- Chapter 3 will discuss prominent related work that explores Large Language Models as agents acting in a situated environment and covers the current state of Multi-agent Coordination Methods and Environments.
- Chapter 4 will introduce the LLM-Coordination Framework to enable LLMs to complete coordination games.
- Chapter 5 will cover the environments we have used to conduct experiments, in-